51–66 Find the absolute maximum and absolute minimum values of f on the given interval. Question#01 from 51–56

51. $f(x) = 12 + 4x - x^2, \quad [0, 5]$

52. $f(x) = 5 + 54x - 2x^3, \quad [0, 4]$

53. $f(x) = 2x^3 - 3x^2 - 12x + 1, \quad [-2, 3]$

54. $f(x) = x^3 - 6x^2 + 5, \quad [-3, 5]$

55. $f(x) = 3x^4 - 4x^3 - 12x^2 + 1, \quad [-2, 3]$

56. $f(t) = (t^2 - 4)^3, \quad [-2, 3]$

29–30 Find the local maximum and minimum values of f using both the First and Second Derivative Tests. Which method do you prefer?

Question#01 from 29–30

29. $f(x) = 1 + 3x^2 - 2x^3$

30. $f(x) = \dfrac{x^2}{x - 1}$

8–70 Find the limit. Use l'Hospital's Rule
If there is a more elementary method, consi
l'Hospital's Rule doesn't apply, explain wh

8. $\lim\limits_{x \to 3} \dfrac{x - 3}{x^2 - 9}$

Question#3 All

9. $\lim\limits_{x \to 4} \dfrac{x^2 - 2x - 8}{x - 4}$

11. $\lim\limits_{x \to 1} \dfrac{x^7 - 1}{x^3 - 1}$

13. $\lim\limits_{x \to \pi/4} \dfrac{\sin x - \cos x}{\tan x - 1}$

15. $\lim\limits_{t \to 0} \dfrac{e^{2t} - 1}{\sin t}$

17. $\lim\limits_{x \to 1} \dfrac{\sin(x - 1)}{x^3 + x - 2}$

19. $\lim\limits_{x \to \infty} \dfrac{\sqrt{x}}{1 + e^x}$

1–8 Evaluate the integral by making the given substitution.

1. $\int \cos 2x\,dx, \quad u = 2x$

2. $\int xe^{-x^2}\,dx, \quad u = -x^2$

3. $\int x^2\sqrt{x^3+1}\,dx, \quad u = x^3 + 1$

4. $\int \sin^2\theta \cos\theta\,d\theta, \quad u = \sin\theta$

5. $\int \dfrac{x^3}{x^4-5}\,dx, \quad u = x^4 - 5$

9–54 Evaluate the indefinite integral.

9. $\int x\sqrt{1-x^2}\,dx$

10. $\int (5-3x)^{10}\,dx$

11. $\int t^3 e^{-t^4}\,dt$

12. $\int \sin t\sqrt{1+\cos t}\,dt$

13. $\int \sin(\pi t/3)\,dt$

14. $\int \sec^2 2\theta\,d\theta$

15. $\int \dfrac{dx}{4x+7}$

16. $\int y^2(4-y^3)^{2/3}\,dy$

17. $\int \dfrac{\cos\theta}{1+\sin\theta}\,d\theta$

18. $\int \dfrac{z^2}{z^3+1}\,dz$

1–4 Evaluate the integral using integration by parts with th
indicated choices of u and dv.

1. $\int xe^{2x}\,dx; \quad u = x, \; dv = e^{2x}\,dx$

2. $\int \sqrt{x}\ln x\,dx; \quad u = \ln x, \; dv = \sqrt{x}\,dx$

3. $\int x\cos 4x\,dx; \quad u = x, \; dv = \cos 4x\,dx$

4. $\int \sin^{-1}x\,dx; \quad u = \sin^{-1}x, \; dv = dx$

Integration by parts

21. $\int e^{3x} \cos x \, dx$

22. $\int e^{x} \sin \pi x \, dx$

EXAMPLE 5 Calculate $\int_0^1 \tan^{-1}x\, dx$.

Integration by parts:definite

Integrals of Powers of:

(a) Sine and Cosine (b) Secant and Tangent

(i) $\int \sin^3 x \cos^2 x \, dx$ (ii) $\int \cos^6 y \sin^3 y \, dy$

(iii) $\int \tan^6 x \sec^4 x \, dx$ (iv) $\int \tan^5 x \sec^7 x \, dx$

Evaluate the Integral by Partial fraction.

(a) Find $\int \frac{dx}{x^2-a^2}$, when $a \neq 0$

(b) Find $\int \frac{x^4-2x^2+4x+1}{x^3-x^2-x+1}\,dx.$

(c) Find $\int \frac{x^3+x}{x-1}\,dx$

(d) Find $\int \frac{x^2+2x-1}{2x^3+3x^2-2x}\,dx.$

→ we already covered these problems in the class.